

A noninvasive model for renal biopsy prediction from retinal images

Corresponding Author: Professor Haotian Lin

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

In this study, Wu and colleagues develop, internally validate, and externally validate a kidney intelligent diagnosis system anchored on retinal imaging applied to CKD screening, pathological diagnosis, and long-term prognosis in patients with CKD. Overall, the authors are very clear in their methodology and present compelling results using high-quality data, including multi-national datasets. The authors also clearly address and highlight where their study fits in the literature and clinical implications, using excellent visuals. However, there are several important issues the authors must address:

Major

1. The authors selected the “5 most common types of CKD” for their prediction modeling, including hypertensive nephropathy. There has been a strong push to move away from the term hypertensive nephropathy, as this is somewhat vague and likely encompasses various, more specific etiologies of injury. Have the authors explored using other, more neutral and descriptive terms such as arteriosclerosis? Given that all adjudication with pathologists and nephrologists have already been completed, this should be feasible
2. Related to the above point, the authors categorize MCD or FSGS as a combined feature. It is reasonable to combine primary MCD and FSGS, with their shared pathological mechanism as podocytopathies along a spectrum of severity. However, in practical terms, secondary FSGS is much more frequently seen and may be helpful to have as its category since the clinical management of patients with primary MCD is significantly different from patients with FSGS
3. Of the 6 clinical features used in the pathological staging model, renal ultrasound measurements include L renal length, clear demarcation of cortex of medulla, and echogenicity of the medulla. Outside of L renal length, I don't see these features listed in Supplemental Table 2. How were these ultrasound features determined? Does this require adjudication, and would this not limit practical implementation of this model?

Minor

1. Beyond the five most common findings on CKD that the authors select, were other pathological findings classified? Or is there a plan to expand this further? Worth noting in future directions if this is the case, or as a limitation – interstitial inflammation? Interstitial fibrosis and tubular atrophy? These are common histological features that have prognostic value.
2. can the authors comment on the severity of their pathological diagnosis prediction? For example, can the authors make any definitive statements on mild/moderate/severe IgA or DM nephropathy, even before looking into the prediction of CKD progression? My understanding is the model was built to determine only the presence of the five pathological categories
3. The authors note in their limitations that the study cohort only included Chinese and Somali patients, but their goal with deploying KIDS on a cloud platform is to promote its use more broadly, with model training on more diverse clinical and demographic cohorts. Would be worth having a comment about its use in the Americas as well, especially to allow for greater diversity among patients of West African ancestry, Hispanic ethnicity, etc

(Remarks on code availability)

Reviewer #2

(Remarks to the Author)

My concerns are as below:

1. Sample size lacks formal power calculations and has concerning issues, particularly in key subgroups like Somalia cohort (n=47) and less common pathological types like MCD/FSGS. This raises questions about statistical power to detect meaningful differences in smaller populations.
2. Missing data handling lacks adequate documentation, with no clear explanation of how missing values were managed, no reporting of missing data percentages, and no sensitivity analyses to assess impact of missing data on results.
3. Model calibration is insufficiently reported, with calibration plots relegated to supplementary materials and no formal calibration testing (such as Hosmer Lemeshow) presented. This limits understanding of models' probability estimation accuracy.
4. Multiple testing concerns remain unaddressed, with no clear adjustment for multiple comparisons across numerous statistical tests performed.
5. Generalizability of results is limited by geographic clustering of validation cohorts and incomplete data in Somalia cohort, raising questions about model performance across truly diverse healthcare settings and populations.

(Remarks on code availability)

N/A

Reviewer #3

(Remarks to the Author)

The paper introduces the Kidney Intelligent Diagnosis System (KIDS), a non-invasive model for diagnosing chronic kidney disease using retinal imaging. This method could be an alternative to the invasive and risk-prone renal biopsy.

The potential of using retinal fundus photographs to detect both ophthalmic and systemic disorders, including CKD, has already been demonstrated in earlier publications. However, the prediction of disease progression and pathological diagnosis, comparison with human performance, and validation in prospective and external cohorts is relevant to the community. Performances of the models are high and show potential for further research. The methodology is sound.

Further validation, as acknowledged by the authors, will be needed to demonstrate broad generalizability of these results.

Some weaknesses remain in the current manuscript, and I would encourage the authors to consider the following points:

- The use of saliency maps and nomograms to enhance interpretability is interesting but could be further developed to show how physicians could use it to challenge KIDS predictions.
- The term "metadata" is confusing as it relates in reality to actual standard "clinical data". I would recommend using another term to avoid confusion.
- The "metadata-only" model and "hybrid" model are compared with the Net Reclassification Index Statistics (NRI) which has been criticized in the literature. These limitations should be discussed by the authors.
- Given the relatively good performance of the metadata-only model, the risk-benefit of having retinal images as part of the model, especially in settings where high-quality images could be challenging to obtain, would be interesting to explore.
- In general, mitigation strategies to address the varying quality of the retinal images (and their impact on the model predictions) should be further discussed.
- The fact that no gold labels (i.e. biopsy-confirmed results) were available for the Somali cohort gives less strength to these results. In addition, it is not clear whether the Somali cohort was labelled by Somali nephrologists (which showed lower relative performance in their diagnostic capabilities) or by other specialists as well.
- How the original model was adapted to "accommodate the existing clinical data in Somalia" is not very clear. Is that what appears in Supplementary Table 3?
- There is an error in the supplementary table numbering.

(Remarks on code availability)

The code is not freely available, as stated in the README file from Github:

"Code and de-identified participant data of KIDS will be provided upon reasonable request by signing the KIDS_Access_Form KIDS_Access_Form.pdf and License_for_KIDS_Code License_for_KIDS_Code.pdf. Please provide appropriate protocols, analysis plans, and data exchange plans and require institutional approval prior to transmitting any information contained in the data. Requests should be directed to the lead corresponding author (linht5@mail.sysu.edu.cn). Submitted license and data access forms will be evaluated by the data manager."

Version 1:

Reviewer comments:

Reviewer #2

(Remarks to the Author)

I have carefully reviewed the revised manuscript and the authors' responses to the reviewers' comments. The authors have appropriately addressed all the concerns raised during the review process.

(Remarks on code availability)

I have carefully reviewed the revised manuscript and the authors' responses to the reviewers' comments. The authors have appropriately addressed all the concerns raised during the review process. The revised manuscript is well-written and demonstrates substantial improvement.

Reviewer #3

(Remarks to the Author)

I thank the authors for their revised submission and for addressing the comments and suggestions of the first round of reviews.

While limitations remain regarding generalizability of the results to a broader population, the additions and clarifications provided by the authors address the concerns raised previously.

However, going through the source code used to produce the results and made available for this second review, I have some concerns:

- The code is undocumented, making it difficult to understand. Besides, some code functions are empty, e.g. `fboxST`, and cannot be tested.
- The fact that model predictions are truncated (by doing operations like $\text{int}(p*100)/100$) makes the model probably less accurate than using rounding functions.
- It looks from the file processing the Kashi data (`data prep/kashi_data cleaning.py`) that the imputation of missing values is performed on the entire dataset, before the split into train, test, and validation. This is problematic as it can lead to data leakage and alter the results of the prediction model. Imputation should be performed on the train, test, and validation sets independently. Can the authors ensure that there is no data leakage?
- In the same file, the imputation using KNN shows that the nearest neighbors are chosen based on a small subset of entire features, which is suboptimal. In addition, binary variables are imputed with the KNN methods, instead of using modes, contrary to what is stated in the supplementary material on page 2. The criteria of 5% of missing data for the use of KNN is not present which is also a contradiction with the text.

(Remarks on code availability)

The code was made available through a Google drive link. The code itself is not commented and does not come with clear instructions or a README.

Some of the data files (source data and model parameters) as well as function bodies needed to run the code are missing and the results can therefore not be reproduced.

Version 2:

Reviewer comments:

Reviewer #3

(Remarks to the Author)

I thank the authors for the revised version of the source code and for providing the additional supplementary information on GitHub and as attachments. The added documentation and clarification successfully address the concerns raised in the previous review round.

Minor Comments:

There remains some room for improvement in the codebase. Specifically, I would recommend the following for future refinement:

- Removal of duplicated functions
- Reorganization of the file structure for clarity and maintainability
- Removal of hard-coded parameters and file paths

(Remarks on code availability)

The code has been made available for review, and documentation was added on the Github repository, in particular through a README.

Missing information from the previous review round was provided, except for the model's parameters which cannot be shared due to intellectual property concerns.

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Reviewer #1 (Remarks to the Author):

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Major

1. The authors selected the “5 most common types of CKD” for their prediction modeling, including hypertensive nephropathy. There has been a strong push to move away from the term hypertensive nephropathy, as this is somewhat vague and likely encompasses various, more specific etiologies of injury. Have the authors explored using other, more neutral and descriptive terms such as arteriosclerosis? Given that all adjudication with pathologists and nephrologists have already been completed, this should be feasible.

R1-1: We sincerely appreciate your insightful suggestion regarding the term “hypertensive nephropathy”. Initially, we used “hypertensive nephropathy” as it has been a conventional term frequently employed in the classification of kidney diseases. However, we acknowledge the growing consensus in recent years advocating for more precise terminology. In light of this valuable feedback, we have revised the expression to “arterionephrosclerosis” to adopt a more neutral and precise term. This change aligns with current literature and pathology textbooks, ensuring consistency with contemporary clinical and academic standards.

2. Related to the above point, the authors categorize MCD or FSGS as a combined feature. It is reasonable to combine primary MCD and FSGS, with their shared pathological mechanism as podocytopathies along a spectrum of severity. However, in practical terms, secondary FSGS is much more frequently seen and may be helpful to have as its category since the clinical management of patients with primary MCD is

significantly different from patients with FSGS.

R1-2: We classified primary minimal change disease (MCD) and primary focal segmental glomerulosclerosis (FSGS) as a combined category due to their shared pathological mechanism as podocytopathies, and their overlapping clinical features and treatment approaches (*Nat Rev Nephrol.* 2016, 12(12): 768-776; *Kidney360.* 2024, 5(8): 1197-1199). Cases lacking identifiable secondary causes but showing pathological evidence of FSGS or diffuse podocyte foot process effacement on electron microscopy were also classified under the primary MCD/FSGS (*Kidney Int. Rep.* 2023, 9(3): 661-670).

It is crucial in the management of patients with CKD to distinguish primary MCD/FSGS from other major pathological types, such as IgA nephropathy (IgAN), membranous nephropathy (MN), diabetic nephropathy (DN), and arterionephrosclerosis (ANS). Moreover, IgAN, MN, primary MCD and FSGS represent the most common types of primary glomerular diseases, each necessitating distinct treatment protocols. While both primary MCD/FSGS and MN are leading causes of nephrotic syndrome, they differ significantly in treatment and prognostic outcomes. This underscores the clinical importance of accurately differentiating among these CKD pathological types. For example, primary MCD/FSGS is thought to be driven by circulating factors that cause podocyte injury and may respond to treatments such as corticosteroids and immunomodulatory agents. On the contrary, DN and ANS do not necessitate immunosuppression therapy. In this study, the KIDS demonstrated to accurately identify the five pathological types mentioned above, with performance comparable to that of invasive renal biopsy. This represents a critical advancement in the non-invasive diagnosis of CKD.

We are conducting a large-scale real-world study, which expands the patient cohort to a broader range of CKD pathological types. This will allow us to refine the diagnostic accuracy and extend the capability of the KIDS system, particularly in identifying critical pathological types associated with secondary FSGS. We sincerely appreciate your insightful comments, which will help us prioritize secondary FSGS in our future study.

3. Of the 6 clinical features used in the pathological staging model, renal ultrasound measurements include L renal length, clear demarcation of cortex of medulla, and echogenicity of the medulla. Outside of L renal length, I don't see these features listed in Supplemental Table 2. How were these ultrasound features determined? Does this require adjudication, and would this not limit practical implementation of this model?

R1-3: We would like to clarify that the pathological staging model is specifically designed to assess whether the percentage of sclerotic glomeruli (PSG) exceeds 75%. The variables listed in Supplemental Table 2, however, correspond to the model used to classify patients into specific pathological types, which is separate from the PSG model.

The six variables used in the PSG model include hemoglobin, creatinine, uric acid, and renal ultrasound measurements (left renal length, clear demarcation of the cortex, and echogenicity of the medulla). The ultrasound characteristics of the patients in our cohort are summarized in Response Table 1, which is also provided in the Supplementary Information as Supplementary Table 7.

Regarding the ultrasound features, these were determined by experienced ultrasonographers who measured kidney size, assessed the cortex-medulla distinction, and evaluated medullary echogenicity. Actually, these ultrasound parameters are commonly used in clinical practice. Given that renal ultrasound is a widely available and routinely performed diagnostic tool, we believe that these features would not limit the practical implementation of our model. For example, even in a resource-limited country such as Somalia, renal ultrasound is becoming increasingly accessible and integrated into clinical practice.

For areas with limited access to ultrasound equipment, we have also developed a simplified version of the model that excludes ultrasound variables. The simplified model, while omitting the ultrasound features, still maintains strong predictive performance with an AUC ranging from 0.641 to 0.802.

Response Table 1. Renal ultrasound features used in pathological diagnosis and staging model.

	Pathological diagnosis and staging model					
	Development set	Internal test set	External test set	Prospective test set	Human-machine comparisons dataset	Real-world dataset
Participants	1818	321	358	301	256	717
Left renal length, cm	10.2 [9.7–10.9]	10.3 [9.6–10.9]	10.5 [10.4–11.0]	10.4 [9.9–10.9]	10.1 [9.8-10.9]	11.1 [10.3–11.5]
Left renal width, cm	5.0 [4.6–5.4]	5.0 [4.6–5.4]	4.9 [4.8–5.8]	5.0 [4.6–5.4]	4.9 [4.6-5.3]	5.1 [4.6–5.4]
Right renal length, cm	10.2 [9.6–10.9]	10.2 [9.6–10.8]	10.3 [10.3–10.5]	10.3 [9.8–10.9]	10.2 [9.8-10.9]	10.9 [10.3–11.2]
Right renal width, cm	4.6 [4.1–5.1]	4.6 [4.2–5.0]	4.8 [4.7–5.8]	4.6 [4.1–4.9]	4.6 [4.3-5.1]	4.8 [4.4–5.1]
Clear demarcation of cortex and medulla	1153 (63.4)	208 (64.8)	338 (94.4)	221 (73.4)	197 (79.1)	561 (90.9)
Echogenicity of the medulla						
No elevated medullary echogenicity	890 (49.0)	160 (49.8)	300 (83.8)	179 (59.5)	153 (61.2)	481 (78.0)
Slightly elevated medullary echogenicity	199 (10.9)	38 (11.8)	13 (3.6)	38 (12.6)	31 (12.4)	40 (6.5)
Elevated medullary echogenicity	729 (40.1)	123 (38.3)	45 (12.6)	84 (27.9)	66 (26.4)	96 (15.6)

Notes: Data are number of individuals (%) or median [IQR]. The renal ultrasound measurements from Somalia (n=100) in the real-world dataset is missing.

Minor

4. Beyond the five most common findings on CKD that the authors select, were other pathological findings classified? Or is there a plan to expand this further? Worth noting in future directions if this is the case, or as a limitation – interstitial inflammation? Interstitial fibrosis and tubular atrophy? These are common histological features that have prognostic value.

R1-4: In our cohort, the number of patients with IgAN (42.00%), MN (22.08%), DN (12.42%), ANS (12.05%), and MCD/FSGS (11.45%), was relatively higher compared to other pathological types. The proportion distribution of the above five pathological types is consistent with several epidemiological surveys of renal biopsies in Chinese population (*Renal Failure*, 2022, 44(1): 987-93; *Postgraduate Medical Journal*. 2023, 21; 100(1179): 20-27; *Scientific Reports*, 2020, 10(1): 10994; *Kidney Diseases*. 2018, 4(1): 10-19). Therefore, we focused on analyzing and constructing AI models for these five pathological types of CKD. Furthermore, we are highly interested in the histological features you mentioned, including interstitial inflammation, interstitial fibrosis, and tubular atrophy. We are currently in the process of collecting these histological data from CKD patients and will analyze their association with patient outcomes in future studies. Additionally, in our ongoing research on the real-world application of the KIDS system, we plan to expand the model's capability to predict a broader range of pathological types, such as interstitial nephritis and lupus nephritis. We look forward to presenting these findings in future study.

In Manuscript (Line 424, Page 16):

“Additionally, we also aim to expand the model's capability to predict a wider range of pathological types. Future studies will also focus on the histological characteristics of CKD to examine their prognostic relevance.”

5. Can the authors comment on the severity of their pathological diagnosis prediction? For example, can the authors make any definitive statements on mild/moderate/severe IgA or DM nephropathy, even before looking into the prediction of CKD progression? My understanding is the model was built to determine only the presence of the five pathological categories.

R1-5: In this study, we developed the KIDS to implement comprehensive process management in CKD diagnosis and treatment decision-making, including (1) detecting CKD in populations, (2) conducting pathological diagnosis and staging in identified CKD patients, and (3) predicting disease prognosis on the basis of the pathological diagnosis.

In the realm of pathological diagnosis, the KIDS system demonstrates high accuracy in identifying the five most prevalent pathological types of CKD, achieving AUCs between 0.794 and 0.942 in a multicenter, multi-ethnic validation study, outperforming the mean nephrologist performance by 26.98% in accuracy. While the current model primarily focuses on categorizing CKD by pathological type, future enhancements could expand its capabilities to assess the severity of CKD, further broadening its clinical application.

To address the question of the severity of CKD, we have integrated an evaluation of the percentage of sclerotic glomeruli (PSG), a well-established indicator of disease severity across various pathological types. Higher degrees of global glomerulosclerosis have consistently been linked to poorer outcomes in CKD patients. Therefore, we developed an AI model for the pathological staging of CKD patients, and its AUC reached 0.901 (95% CI: 0.847, 0.955) in the internal test dataset for identifying patients with high PSG without renal biopsy. When evaluated via prospective and external tests, the KIDS achieved similar AUCs of 0.883 (95% CI: 0.813, 0.954) and 0.919 (95% CI: 0.881, 0.958), respectively.

Indeed, the ability to predict disease stages (mild, moderate, or severe) for each pathological type would be a significant breakthrough of CKD diagnosis. In our large-scale real-world study, which will enroll a cohort of 50,000 to 100,000 patients, we plan to investigate disease severity stratification within each pathological type of CKD. This will enable us to offer more accurate prediction regarding disease progression.

6. The authors note in their limitations that the study cohort only included Chinese and Somali patients, but their goal with deploying KIDS on a cloud platform is to promote its use more broadly, with model training on more diverse clinical and demographic cohorts. Would be worth having a comment about its use in the Americas as well,

especially to allow for greater diversity among patients of West African ancestry, Hispanic ethnicity, etc

R1-6: While the current study cohort primarily consists of Chinese and Somali patients, we would like to emphasize that the KIDS system has been developed for broad applicability and is already deployed on a cloud platform to facilitate wider application. We are committed to expanding the diversity of clinical and demographic cohorts for model training, which will enhance its generalizability. We are currently enrolling Portuguese, British and Thai patients in our real-world study.

In response to your comment, we recognize the importance of including a broader range of populations, particularly those in the Americas, with diverse ethnic backgrounds such as West African and Hispanic ancestry. We plan to incorporate these populations in future studies, further refining and validating the KIDS system to ensure its effectiveness across diverse patient groups.

In Manuscript (Line 653, Page 16):

“Finally, because our study cohort included Chinese and Somali patients, the generalizability in a wider range of countries and ethnicities will need to be validated. To enhance the clinical utility of the KIDS system across more diverse regions and populations, we have implemented its deployment on a cloud platform (<https://zockids.gzzoc.com/modelstore/#!/model/list>) enabling broader accessibility and application.”

Reviewer #2 (Remarks to the Author):

My concerns as below:

1. Sample size lacks formal power calculations and has concerning issues, particularly in key subgroups like Somalia cohort (n=47) and less common pathological types like MCD/FSGS. This raises questions about statistical power to detect meaningful differences in smaller populations.

R2-1: The pathological diagnosis models were developed using data from CKD patients who underwent renal biopsy, ensuring accurate and reliable predictions of pathological types. To ensure robust validation of the models in both the prospective and external

test cohorts, the required sample sizes were determined by a statistician based on the distribution of pathological types and the model performance observed in the internal test dataset. With a power of 0.8 and an alpha level of 0.05, the minimum sample size for each cohort was calculated to be 180 CKD patients. In this study, the sample sizes for both the prospective and external validation cohorts exceeded this minimum requirement, with 301 patients included in the prospective validation cohort and 498 patients from four hospitals in Guangdong Province, Guangxi Zhuang Autonomous Region, and Xinjiang Uyghur Autonomous Region, China, for external validation. These sample sizes were adequate to ensure statistical power and provided sufficient representation for a comprehensive validation of the model's performance across diverse clinical settings.

In Somalia, due to constraints in healthcare infrastructure and economic conditions, renal biopsy is scarcely available, and clinical records for hospitalized patients are often incomplete. As a result, only a limited number of CKD patients have access to comprehensive diagnostic data. To overcome this challenge, we developed a simplified pathological diagnosis model specifically adapted to the clinical data available in Somalia, ensuring it remains both relevant and applicable to the local healthcare context.

Thank you for the reviewer's suggestion. To improve the model's applicability, we expanded the validation cohort to include patients from multiple regions and ethnic groups. Our model demonstrated excellent generalization ability in this expanded real-world dataset, further supporting its potential for broader clinical application across diverse populations.

The expanded validation dataset in the revision includes the following updates:

1. Expansion of the Xinjiang External Validation Cohort: We have further incorporated CKD patients who underwent renal biopsy at the First People's Hospital of Kashi (FPHK) in Kashi, Xinjiang Uygur Autonomous Region, China. The cohort size has increased from 140 to 272 patients.

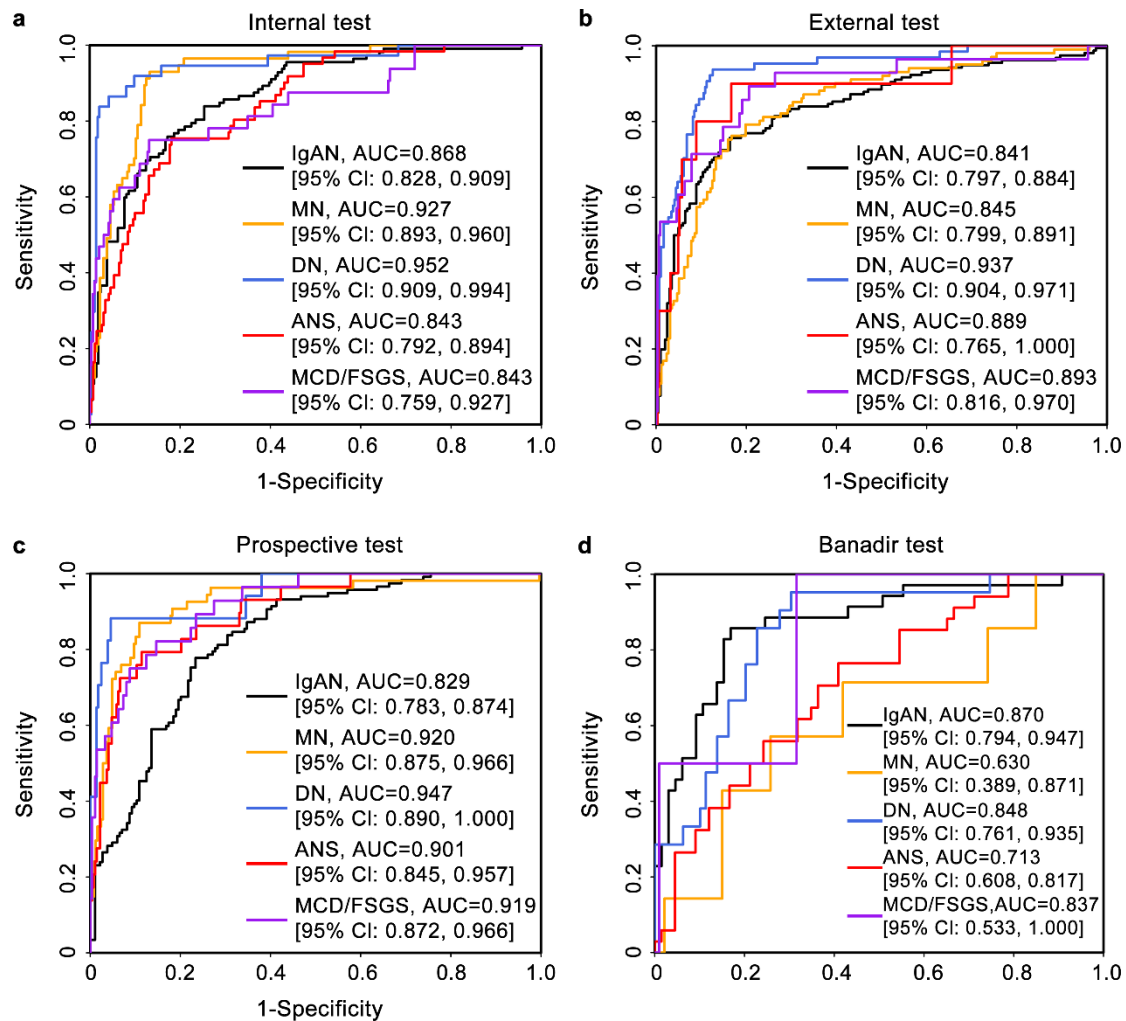
2. New External Validation Cohorts: We added two new external validation datasets from central China: Shanxi Provincial Traditional Chinese Medicine Institute (SPTCMI) in Taiyuan City, Shanxi Province, and The Second Affiliated Hospital of

Xi'an Jiaotong University (SAH) in Xi'an City, Shaanxi Province, with a combined total of 215 patients.

3. Expansion of the Somalia External Validation Cohort: We have made every effort to expand the cohort in Somalia from 47 to 99 patients, including cases of IgA nephropathy (n=35), membranous nephropathy (n=7), diabetic nephropathy (n=21), arterionephrosclerosis (n=34), and minimal change disease/focal segmental glomerulosclerosis (n=2).

These enhancements significantly improve the diversity and representativeness of the validation cohorts, further displaying the model's generalizability and applicability across different populations and regions.

While the patient cohort from Somalia is relatively small compared to other external validation datasets and our expanded real-world dataset, the simplified pathological diagnosis model has demonstrated promising performance in the Somalia cohort (Supplementary Figure 5). Due to the lack of renal biopsy capabilities in Somalia, the good performance of the KIDS system in predicting pathological diagnoses has been well received by local nephrologists. We are encouraged by this recognition and look forward to continuing support for KIDS in real-world applications. As the dataset grows, we anticipate further improvements in the model's performance, enabling more reliable outcomes.



Supplementary Figure 5. Performance of the simplified clinical data model for pathological diagnosis in patients from different countries and ethnic groups. ROC curves showing the performance for the pathological diagnosis of IgAN, MN, DN, ANS, and MCD/FSGS (a-d) in the internal test set, external test set, prospective test set and Banadir hospital of Somalia test set. DN, diabetic nephropathy; IgAN, IgA nephropathy; MN, idiopathic membranous nephropathy; ANS, arterionephrosclerosis; MCD/FSGS, idiopathic minimal change disease and focal segmental glomerulosclerosis. AUC, area under the curve; CI, confidence interval.

2. Missing data handling lacks adequate documentation, with no clear explanation of how missing values were managed, no reporting of missing data percentages, and no sensitivity analyses to assess impact of missing data on results.

R2-2: The missing data were carefully managed during the data preprocessing, with

guidance from a statistician. In the development dataset, variables that were not commonly available in the external test sets were excluded. Samples with more than 25% of missing values were removed. Specifically, the BMI variable had 20% missing values, the medical history had less than 0.3% missing, laboratory tests had missing values ranging from 1% to 3%, and kidney ultrasound measurements had approximately 5% missing values. Similar patterns were observed in the other datasets, except for kidney ultrasound measurements and BMI, where missing data ranged from 10% to 30%, and 3% to 10%, respectively.

For imputation, binary variables were imputed with the mode, while continuous variables were imputed using the median or KNN imputation with $k=3$, particularly when the missing data exceeded 5%. We have provided a detailed description of these procedures in the data preprocessing section of the Supplementary Information. The sample size was determined by a statistician, taking into account the minimal amount of missing data for most variables. As a result, we believe sensitivity analyses may not be required in this case.

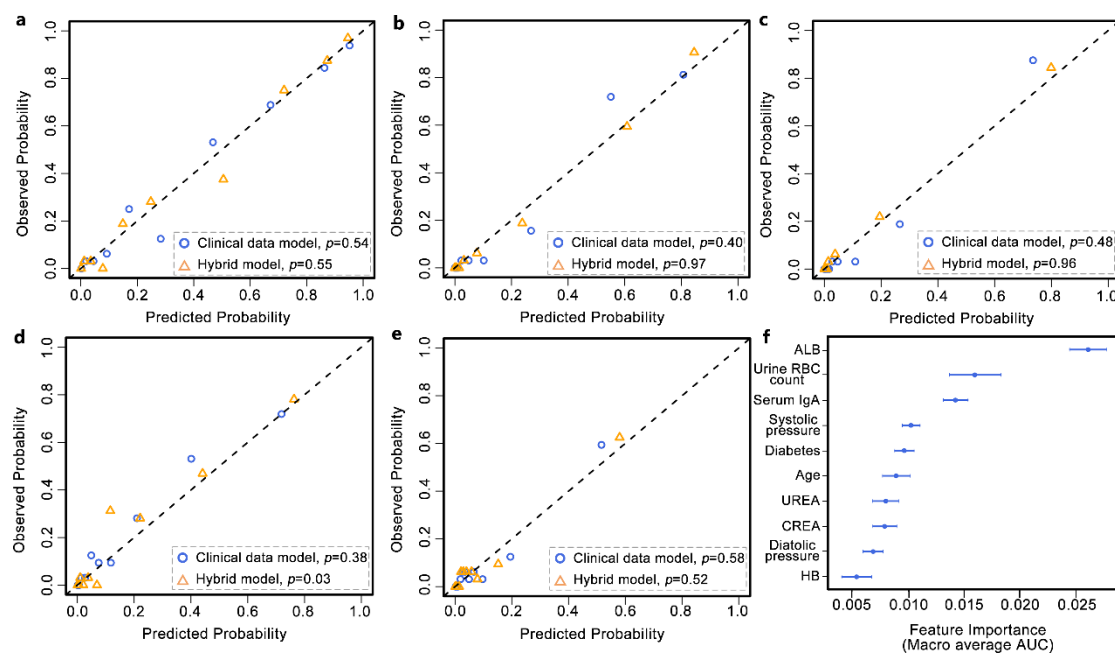
In Supplementary Information (Line 39, Page 2):

“Clinical data preprocessing. First, categorical variables were transformed into dummy variables for analysis. Missing data were then addressed. In the development dataset, variables with more than 25% missing values or those unavailable in the external validation centers, were excluded. Samples with over 25% missing data were also removed. The BMI variable had 20% missing values, the history of disease variable had less than 0.3% missing values, laboratory test data were missing in approximately 1-3% of cases, and kidney ultrasound measurements had about 5% missing values. Similar trends were observed in the other datasets, with the exception that kidney ultrasound data had higher missing rates (approximately 10-30%) and BMI had fewer missing values (3-10%). For imputation, binary variables were filled with the mode, and continuous variables were imputed using the median or K-nearest neighbor (KNN) imputation with $k=3$ when more than 5% of data were missing. Subsequently, data from the FAH retrospective cohort were randomly split into three mutually exclusive groups: training (75%), validation (10%), and internal test (15%) sets. Finally, continuous

variables were standardized to have a zero mean and unit variance, with the means and standard deviations from the training set applied to standardize the validation and test sets.”

3. Model calibration is insufficiently reported, with calibration plots relegated to supplementary materials and no formal calibration testing (such as Hosmer Lemeshow) presented. This limits understanding of models’ probability estimation accuracy.

R2-3: We have included the test results as requested. Specifically, we performed the Hosmer-Lemeshow goodness of fit tests and have supplied the results along with the calibration plots in Supplementary Figure 3. As with other tests, the null hypothesis (H0) was favored in the Hosmer-Lemeshow test, and thus, the P-values reported on the calibration plots were not adjusted for multiple comparisons.



Supplementary Figure 3. Calibration plots for assessing accuracy and reclassification improvement in predicting pathological diagnosis by adding retinal images

Comparing the predictive probabilities from the clinical data model, hybrid models and the observed probabilities in predicting (a-e) IgAN, MN, DN, ANS and MCD/FSGS. The x-axis represents the probability predicted by the clinical data-only and hybrid

models, and the y-axis represents the observed probability in the pathological diagnosis dataset. Calibration was assessed using the Hosmer-Lemeshow goodness-of-fit test, with P greater than 0.05 indicating consistent calibration with the model. (f) Feature importance of the top 10 variables in the metadata model. ALB, albumin; RBC, red blood cell; serum IgA, serum immunoglobulin A level; UREA, urea, serum; CREA, creatinine, serum; HB, hemoglobin.

To clarify this in the manuscript, we have added the following statement to the Statistical Analysis section *in the Manuscript (Line 201, Page 8)*:

“Compared with those of the clinical data-only model, the AUCs for the hybrid models were greater, but this difference was not statistically significant after applying the Bonferroni correction. The calibration plots and Hosmer-Lemeshow goodness-of-fit tests indicated a good agreement between the observed probabilities and those predicted by both the clinical data-only and hybrid models, with the exception of the hybrid model for ANS (Supplementary Figure 3).”

4. Multiple testing concerns remain unaddressed, with no clear adjustment for multiple comparisons across numerous statistical tests performed.

R2-4: We have made the adjustment accordingly. The Bonferroni correction method was adopted for multiple AUC comparisons. The P-values from the comparisons between the AUCs of the clinical data-only model and the hybrid models have been adjusted using the Bonferroni correction. We have made the corresponding revisions in the manuscript *(Line 201, Page 8)*:

“Compared with those of the clinical data-only model, the AUCs for the hybrid models were greater, but not statistically significant after Bonferroni correction. The calibration plots and Hosmer-Lemeshow goodness-of-fit tests indicated a good agreement between the observed probabilities and those predicted by both the clinical data-only and hybrid models, with the exception of the hybrid model for ANS (Supplementary Figure 3).”

Regarding the IDI analysis, we believe that the confidence intervals alone provide sufficient information, so we have removed the phrase “all of which were statistically

significant” from the same section.

We hope this revision addresses your concerns and improves the clarity of the statistical analysis.

5. Generalizability of results is limited by geographic clustering of validation cohorts and incomplete data in Somalia cohort, raising questions about model performance across truly diverse healthcare settings and populations.

R2-5: We have made great efforts to improve and validate the model generalizability. To further assess the model’s diagnostic capability across different regions, we conducted external validation of CKD patients from 3 independent retrospectively collected datasets: (1) Zhongshan City People’s Hospital [ZPH, Zhongshan City, China]; (2) First People’s Hospital of Foshan [FPH, Foshan City, China]; and (3) Affiliated Hospital of Youjiang Medical University for Nationalities [AHY, Youjiang City, China]. These hospitals are located in 3 cities across 2 provinces in southern China. For external testing in a transnational and multiracial real-world dataset, we further collected 2 datasets: (1) First People’s Hospital of Kashi [FPHK, Kashi, China]; and (2) Banadir Hospital [BH, Mogadishu, Somalia].

To further enhance the model’s applicability and generalizability, we have expanded our validation cohort to include a broader range of regions and ethnic groups. The current external and real-world validation datasets consist of data from seven hospitals in two countries, covering a total of seven ethnicities:

China: Six hospitals from various regions (northern, central, and southern China; geographic distribution detailed in Response Figure 2), with patient cohorts from seven ethnic groups: Han, Uygur, Zhuang, Miao, Tibetan, Yao, and Bouyei.

Somalia: We have doubled the patient enrollment in Somalia, and the current cohort includes patients from five pathological types required for the model validation.

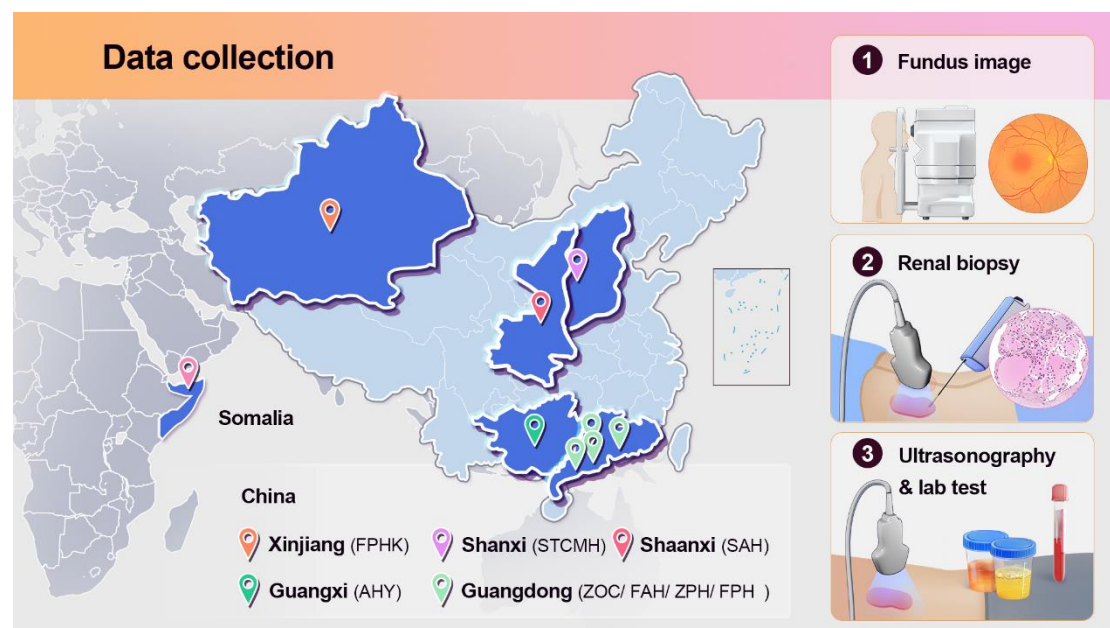
This multicenter and multi-ethnic expansion strengthens the model’s validation and significantly increases its potential for clinical application across diverse populations.

1. Expansion of the Xinjiang External Validation Cohort: We have further incorporated CKD patients who underwent renal biopsy at the First People’s Hospital

of Kashi (FPHK) in Kashi, Xinjiang Uygur Autonomous Region, China. The cohort size has increased from 140 to 272 patients.

2. New External Validation Cohorts: We added two new external validation sites from central China: Shanxi Provincial Traditional Chinese Medicine Institute (SPTCMI) in Taiyuan City and The Second Affiliated Hospital of Xi'an Jiaotong University (SAH) in Xi'an City, with a combined total of 215 patients.

3. Expansion of the Somalia External Validation Cohort: We have made every effort to expand the cohort in Somalia from 47 to 99 patients, including cases of IgA nephropathy (n=35), membranous nephropathy (n=7), diabetic nephropathy (n=21), arterionephrosclerosis (n=34), and minimal change disease/focal segmental glomerulosclerosis (n=2). We would consider this small sample size as a limitation.

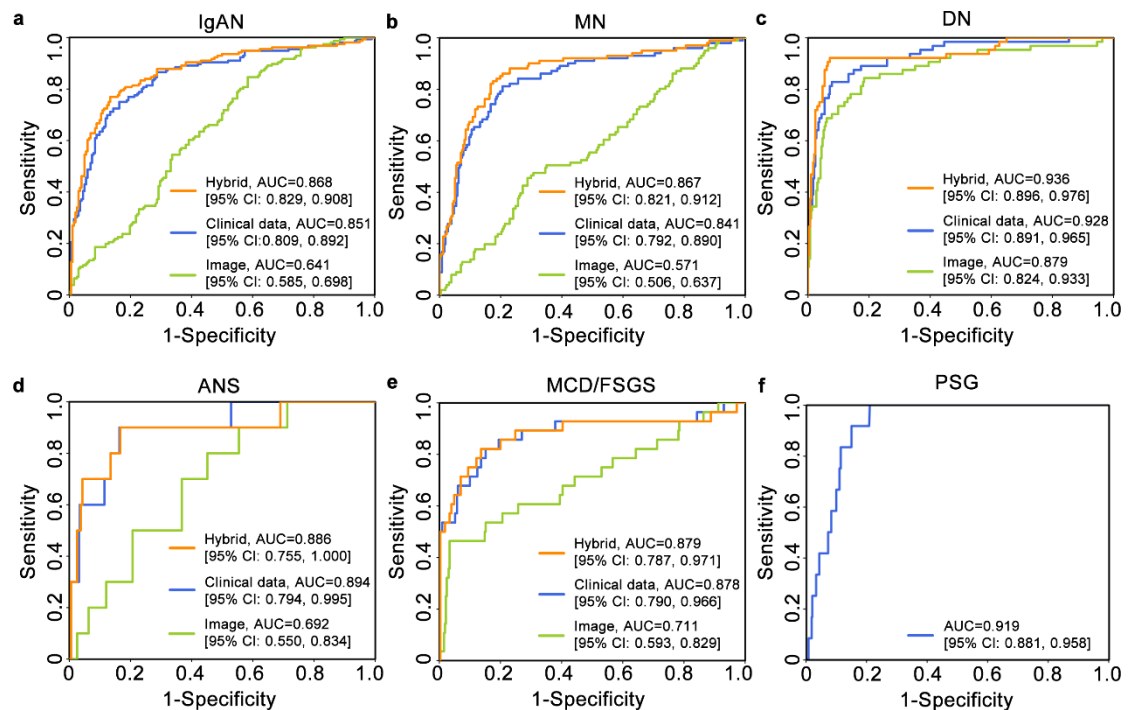


Response Figure 2. External and real-world validation datasets include data from two countries and seven hospitals, representing a diverse cohort across seven ethnic groups. (1) China: **Northern China:** First People's Hospital of Kashi (FPHK), Kashi, Xinjiang Uygur Autonomous Region; **Central China:** Shanxi Provincial Traditional Chinese Medicine Institute (SPTCMI), Taiyuan City, and The Second Affiliated Hospital of Xi'an Jiaotong University (SAH), Xi'an City; **Southern China:** Zhongshan City People's Hospital (ZPH), Zhongshan, Guangdong; First People's Hospital of Foshan (FPH), Foshan, Guangdong; Affiliated Hospital of Youjiang Medical University

for Nationalities (AHY), Youjiang City, Guangxi.. (2) Somalia: Banadir Hospital (BH).

Model performance in external test and multicenter, multi-ethnic real-world validation datasets. The pathological diagnosis model and glomerulosclerosis prediction model both demonstrated strong performance in external test cohorts and multicenter, multi-ethnic validation datasets. These models exhibited satisfactory accuracy across diverse populations, including Han, Zhuang, and Uygur ethnicities in China, as well as Somali patients

In the external test dataset, the KIDS achieved AUCs of 0.867 – 0.936 for all 5 pathological types in the hybrid model (Response Figure 3). The AI model for the pathological staging of CKD patients for the KIDS achieved AUC of 0.919 (95% CI: 0.881, 0.958).

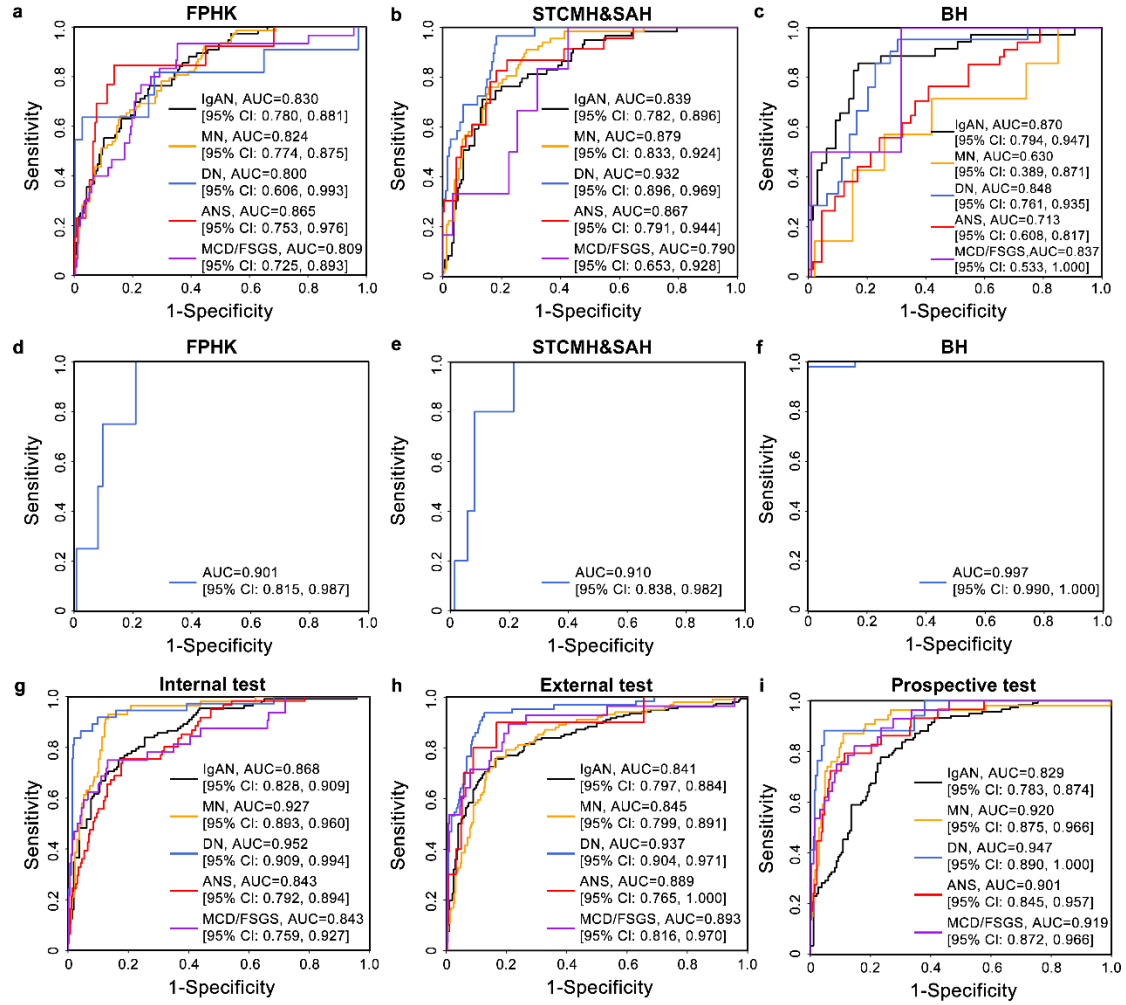


Response Figure 3. Performance of the pathological diagnosis and PSG staging AI models in external test. ROC curves for the performance of pathological diagnosis of IgAN, MN, DN, ANS, MCD/FSGS (a-e), and PSG (f) in the external test set. DN, diabetic nephropathy; IgAN, IgA nephropathy; MN, idiopathic membranous nephropathy; DN, diabetic nephropathy; ANS, arterionephrosclerosis; MCD/FSGS, idiopathic minimal change disease and focal segmental glomerulosclerosis; PSG, the

percentage of the sclerotic glomerulus; AUC, area under the curve; CI, confidence interval.

In the pathological staging model, glomerular lesions with $\geq 75\%$ sclerotic glomeruli are classified as having high PSG. The model output represents the probability of a PSG score $\geq 75\%$. To further assess the diagnostic performance of the model across different regions, we expanded the validation to include multicenter and multi-ethnic real-world datasets from Kashi, Taiyuan, and Xi'an. The AUCs for the pathological types of IgAN, MN, DN, ANS, and MCD/FSGS ranged from 0.790 to 0.932 (Response Figure 4a-b). The pathological staging model demonstrated robust performance, with AUCs of 0.901 (95% CI: 0.815, 0.987) and 0.910 (95% CI: 0.838, 0.912), indicating its reliable ability to differentiate between different stages (Response Figure 4d-e).

Given the limitations in healthcare infrastructure and the economic constraints in Somalia, where renal biopsy is not feasible, we focused on validating the simplified pathological diagnosis model using patient data. Nephrology expert diagnoses served as the reference standard. The simplified pathological diagnosis model achieved AUCs between 0.630 and 0.870, and the PSG staging model showed exceptional performance with an AUC of 0.997 in the Somali dataset (Response Figure 4c, f). Additionally, the model was validated across internal, external, and prospective datasets, demonstrating consistent diagnostic accuracy with AUC values ranging from 0.843 to 0.952 (Response Figure 4g-i), further confirming the model's robustness and clinical applicability.



Response Figure 4. Performance of the Pathological Diagnosis and PSG Staging AI Models in Multicenter and Multi-Ethnic Real-World Datasets from China (Kashi, Taiyuan, and Xi'an) and Somalia. ROC curves are presented for the pathological diagnosis model (a-c), PSG staging model (d-f), and the simplified pathological diagnosis model (g-i). The validation datasets include data from the following institutions: FPHK (First People's Hospital of Kashi, Kashi, Xinjiang Uygur Autonomous Region), SPTCMI (Shanxi Provincial Traditional Chinese Medicine Institute, Taiyuan City), SAH (The Second Affiliated Hospital of Xi'an Jiaotong University, Xi'an City), and BH (Banadir Hospital, Somalia). IgAN, IgA nephropathy; MN, idiopathic membranous nephropathy; DN, diabetic nephropathy; ANS, arterionephrosclerosis; MCD/FSGS, idiopathic minimal change disease and focal segmental glomerulosclerosis; PSG, the percentage of the sclerotic glomerulus; AUC, area under the curve; CI, confidence interval.

Reviewer #3 (Remarks to the Author):

The paper introduces the Kidney Intelligent Diagnosis System (KIDS), a non-invasive model for diagnosing chronic kidney disease using retinal imaging. This method could be an alternative to the invasive and risk-prone renal biopsy.

The potential of using retinal fundus photographs to detect both ophthalmic and systemic disorders, including CKD, as already been demonstrated in earlier publications. However, the prediction of disease progression and pathological diagnosis, comparison with human performance, and validation in prospective and external cohorts is relevant to the community. Performances of the models are high and show potential for further research. The methodology is sound.

Further validation, as acknowledged by the authors, will be needed to demonstrate broad generalizability of these results.

Some weaknesses remain in the current manuscript, and I would encourage the authors to consider the following points:

1. The use of saliency maps and nomograms to enhance interpretability is interesting but could be further developed to show how physicians could use it to challenge KIDS predictions.

R3-1: Thank you for your insightful suggestion. We agree that the use of saliency maps and nomograms to enhance the interpretability of the model is an important aspect and can be further refined for clinical application.

Saliency maps are used to highlight regions of interest (ROIs) in the model's input that are crucial for making predictions, thereby improving interpretability. By emphasizing these areas in retinal images, saliency maps help ensure the model focuses on clinically relevant features. This process also enhances transparency, showing which aspects of the images, the model uses for its predictions, ultimately supporting efforts to refine and improve model accuracy.

Nomograms, in contrast, are designed as practical tools to assist clinicians in making evidence-based predictions about a patient's risk of progression to kidney-related outcomes at specific time points (e.g., 1, 3, or 5 years). The nomogram displays

each contributing factor and be quantified as points, allowing clinicians to visualize the collective risk score. This transparency ensures that clinicians can make informed decisions regarding individualized patient management strategies, whether opting for conservative monitoring or more aggressive interventions such as corticosteroid therapy.

Example:

A 28-year-old male patient diagnosed with IgA nephropathy via renal biopsy was predicted to be at high risk of CKD progression. Using the nomogram, clinicians can assess the patient's risk at multiple time points as follows:

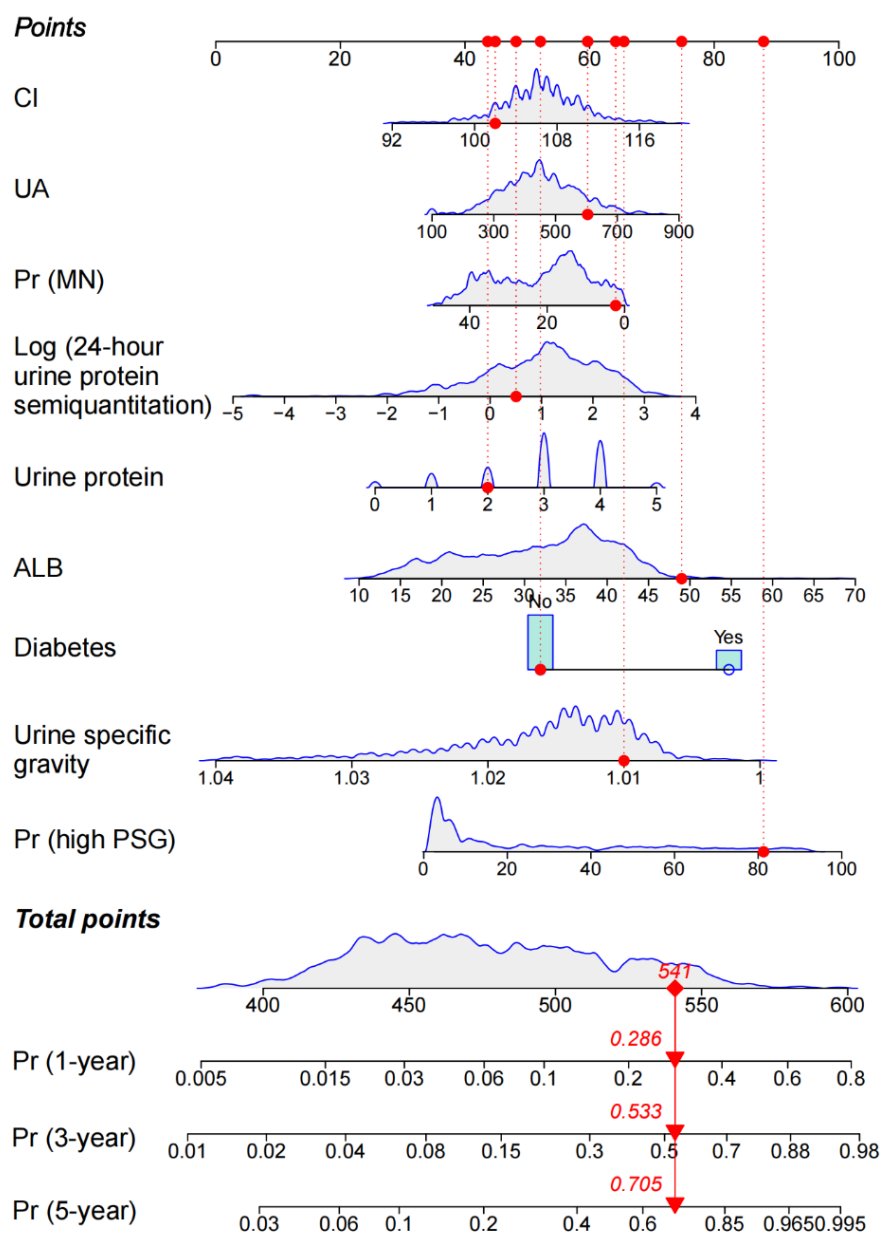
Step 1: Draw vertical lines from each predictor value (e.g., age, proteinuria, predicted pathological type) to the corresponding points on the nomogram scale.

Step 2: Sum the total points (e.g., 541) and plot them on the "Total Points" axis.

Step 3: From the total points, draw a line to the hazard axis to derive the probabilities of progression: 28.6% at 1 year, 53.3% at 3 years, and 70.5% at 5 years.

This visual representation of the prediction allows clinicians to verify the model's output and plan appropriate interventions to mitigate the patient's risk. By integrating tools like saliency maps and nomograms, physicians are empowered to engage critically with AI predictions and tailor treatment strategies accordingly.

We believe that these features not only enhance the interpretability of the model but also empower clinicians to use the predictions to inform their clinical judgment. By visualizing the contributions of each factor, physicians can better understand the model's reasoning, and if necessary, challenge or refine the predictions to optimize patient management.



Response Figure 5. An example of the nomogram using the predicted probability of pathological type and high PSG. The patient was a 28-year-old male patient who was diagnosed with IgA nephropathy following renal biopsy and who subsequently initiated hemodialysis due to disease progression (39 months after administration). Red lines and dots are drawn upward to determine the points received by each variable; the sum (541) of these points is located on the total points axis, and a line is drawn downward to the hazard axes to determine the probability of kidney progression at 1 year (28.6%),

3 years (53.3%) and 5 years (70.5%).

2. The term “metadata” is confusing as it relates in reality to actual standard “clinical data”. I would recommend using another term to avoid confusion.

R3-2: Thank you for your valuable feedback regarding “metadata.” We have replaced “metadata” with “clinical data” accordingly.

3. The “metadata-only” model and “hybrid” model are compared with the Net Reclassification Index Statistics (NRI) which has been criticized in the literature. These limitations should be discussed by the authors.

R3-3: We acknowledge the concerns regarding the Net Reclassification Index (NRI) raised in the literature. Pepe et al. have cautioned against its use due to potential issues with poorly fitted models (*Stat Biosci.* 2015, 7(2): 282-295). However, prior studies have indicated that NRI remains a valid measure when applied to well-calibrated models (*Stat Med.* 2013, 32(9): 1467-82). In our study, calibration plots and Hosmer-Lemeshow goodness-of-fit tests (Supplementary Figure 3) confirm that the models exhibit strong calibration. Therefore, we consider the use of NRI appropriate in this context, as it provides meaningful insights into model reclassification performance following calibration.

We have discussed the limitations and advantages of NRI *in the manuscript (Line 351, Page 13)*:

“Although NRI has faced concerns over poorly fitted models, NRI can still be used when the models are well-calibrated, in this study, our analysis, including calibration plots and Hosmer-Lemeshow goodness-of-fit tests, demonstrates that the models exhibit strong calibration. Herein, the use of NRI is appropriate in this context, as it provides valuable insight into model reclassification after calibration.”

4. Given the relatively good performance of the metadata-only model, the risk-benefit of having retinal images as part of the model, especially in settings where high-quality images could be challenging to obtain, would be interesting to explore.

R3-4: We appreciate your observation regarding the good performance of the clinical data-only model. Indeed, the clinical data-only model has demonstrated good predictive

capability. However, incorporating retinal images has been shown to further enhance the model's AUC in most cases. More importantly, the calibration plots, Hosmer-Lemeshow goodness-of-fit-tests, and NRI results all indicate that retinal images contribute significantly to improving the model's predictive accuracy.

It is important to emphasize that retinal images are not a prerequisite for using the KIDS system. The model is flexible in that patients can choose to rely on the clinical data-only model if retinal images are unavailable. Moreover, the retinal images used in our model represent a cost-effective and accessible form of fundus photography that is commonly used in routine ophthalmologic exams, with a cost of only 47 RMB (approximately 6 USD). This makes the inclusion of retinal images a feasible option for many patients, without significant additional burden.

For patients with conditions such as arterionephrosclerosis (ANS) and diabetic nephropathy (DN), which are associated with retinal changes, we strongly advocate for early retinal screening. Fundus photography can help identify retinal pathologies, allowing for early ophthalmic intervention and ultimately improving long-term visual outcomes.

5. In general, mitigation strategies to address the varying quality of the retinal images (and their impact on the model predictions) should be further discussed.

R3-5: We recognize that variations in retinal image quality may affect model performance. To mitigate this, we applied a series of preprocessing steps to standardize and enhance image quality across datasets from different retinal cameras. This included noise removal, cropping of irrelevant background, and normalization of image size and resolution. Additionally, pixel values were standardized with sample-wise means and standard deviations, ensuring the preservation of clinically relevant features. The strategies employed to enhance image quality are thoroughly detailed in the Supplementary Information.

In Supplementary Information (Line 26, Page 1):

Image preprocessing. With raw retinal images from different retina cameras across datasets, image preprocessing was required to enhance image quality and facilitate model learning. First, noise was excluded. The characters on the background of an

image were removed, the redundant background on the sides was cropped, and only the rectangle containing a retinal region was retained. The images were subsequently normalized to the same size and resolution. The top and bottom of an image were cropped so that the ratio of width to height was 6:5 (the maximum of the ratios for images used by chronic kidney disease (CKD) pathological diagnosis models), and the image was then resized to the same resolution. Furthermore, image augmentation and standardization were adopted via the ImageDataGenerator class from Keras. The images in the training dataset were augmented randomly via brightness adjustment, horizontal flipping, and height and width shifts. Pixel values of all images were standardized with samplewise means and standard deviations.

6. The fact that no gold labels (i.e. biopsy-confirmed results) were available for the Somali cohort gives less strength to these results. In addition, it is not clear whether the Somali cohort was labelled by Somali nephrologists (which showed lower relative performance in their diagnostic capabilities) or by other specialists as well.

R3-6: For patients in the training and validation cohorts with renal biopsy results, the methodology for confirming the pathological diagnosis labels has been described in detail in the Supplementary Methods section. Regarding the Somali cohort, due to limitations in healthcare infrastructure and economic conditions, renal biopsy is not widely available. The pathological diagnoses for Somali patients were made by a panel of senior nephrologists and nephropathologists from China. These experts, all with extensive experience, determined the diagnoses based on comprehensive clinical assessments, laboratory test results, and other available diagnostic data. This approach ensures that the diagnoses are as reliable as possible within the constraints of the available resources.

We have updated the Supplementary Methods to include this clarification, which provides a transparent and thorough explanation of the diagnostic labeling process. This revision aims to address concerns about the reliability of the labels and provide additional context on the challenges faced in resource-limited settings like Somalia.

In Supplementary Methods (Line 9, Page 1):

“In Somalia, due to the challenges posed by limited healthcare infrastructure and

economic conditions, renal biopsy is not routinely available. As such, pathological diagnoses for the Somali cohort were determined through a consensus approach involving a team of senior nephrologists and nephropathologists from China. These experts relied on the available clinical and laboratory data to provide the most accurate possible diagnosis, ensuring that the labels assigned were as reliable as the circumstances permitted.”

7. How the original model was adapted to “accommodate the existing clinical data in Somalia” is not very clear. Is that what appears in Supplementary Table 3?

R3-7: To address the challenges posed by Somalia’s relatively limited healthcare and economic infrastructure, we adapted the original model by streamlining the set of variables required for pathological diagnosis prediction. This adjustment aimed to retain the model’s predictive accuracy while aligning with the practical constraints of the local healthcare setting. The specific variables used in the simplified clinical data model are outlined in Supplementary Table 3.

8. There is an error in the supplementary table numbering.

R3-8: Thank you for your kind reminder. We have revised accordingly.

Reviewer #3 (Remarks on code availability):

The code is not freely available, as stated in the README file from Github:

Given the study involves participant private data, it cannot be disclosed directly. But as we stated in README file from Github, the de-identified participant data and code will be made available upon reasonable request to the corresponding author.

For the needs of the review process, we have uploaded the code package to Google Drive. Please find the link below for anonymous access:

https://drive.google.com/file/d/1N_cV_1bo25uDcTGZJk4LKBIDSCUEO_6U/view?usp=sharing

REVIEWER COMMENTS

Reviewer #3 (Remarks to the Author):

I thank the authors for their revised submission and for addressing the comments and suggestions of the first round of reviews. While limitations remain regarding generalizability of the results to a broader population, the additions and clarifications provided by the authors address the concerns raised previously.

However, going through the source code used to produce the results and made available for this second review, I have some concerns:

1. The code is undocumented, making it difficult to understand. Besides, some code functions are empty, e.g. `fboxST`, and cannot be tested.

R3-1: Thank you for your kind reminder. According to your comment, we have added more code comments and provided code documentation. Additionally, the code functions have been provided. The updated code has been submitted as an attachment and is also accessible via Google Drive at:

<https://drive.google.com/file/d/1MNRjEE1kbOV4S6inqy2BKEn1897cdvX5/view?usp=sharing>.

2. The fact that model predictions are truncated (by doing operations like `int(p*100)/100`) makes the model probably less accurate than using rounding functions.

R3-2: Thank you for pointing out. The truncation was only done for online model deployment, and did not affect the results in the manuscript, which were derived directly from the original model predictions. In accordance with your suggestion, we have updated the code to use standard rounding to three decimal places to preserve precision.

3. It looks from the file processing the Kashi data (`data prep/kashi_data cleaning.py`) that the imputation of missing values is performed on the entire dataset, before the split into train, test, and validation. This is problematic as it can lead to data leakage and alter the results of the prediction model. Imputation should be performed on the train, test, and validation sets independently. Can the authors ensure that there is no data leakage?

R3-3-1: Thank you for your detailed and thoughtful comments. The imputation of missing values wasn't performed on the entire dataset. The imputation was done after the dataset was randomly split into training, validation, and test sets with comparable distributions. At first, the KNN imputer was fit and the modes for binary variables and medians for the continuous variables were calculated on the training dataset; then the KNN imputer, modes, and median were applied to the training, validation, and test dataset.

In the same file, the imputation using KNN shows that the nearest neighbors are chosen

based on a small subset of entire features, which is suboptimal. In addition, binary variables are imputed with the KNN methods, instead of using modes, contrary to what is stated in the supplementary material on page 2. The criteria of 5% of missing data for the use of KNN is not present which is also a contradiction with the text.

R3-3-2: We believe there may have been a misunderstanding regarding the imputation of the Kashi dataset. The Kashi dataset serves solely as an external test set. Upon comparison with the training dataset, we observed substantial differences in feature distributions. Therefore, missing value imputation was performed on the entire Kashi dataset independently.

In the Kashi dataset, binary variables are complete and therefore not subject to imputation. Among the variables, only BMI and kidney ultrasound measurements had more than 5% missingness and were imputed accordingly. Specifically, BMI was imputed using the KNN method, based on clinically relevant and fully observed features-Age, Sex, Diabetes, Primary Hypertension, and Creatinine. These supporting features were not modified during the imputation process; their inclusion was due to a simplified (or “lazy”) coding approach and did not result in any actual changes by the KNN algorithm. Similarly, kidney ultrasound measurements were imputed using KNN, drawing upon a complete and relevant subset of features including Age, Sex, BMI, Hemoglobin, and Creatinine.

To be clear, we have updated the code to reflect that only BMI and kidney ultrasound measurements are imputed using the KNN method. In our implementation, each imputed variable with KNN method is based on a targeted subset of clinically relevant and correlated features, rather than the entire feature set. This design choice aligns with the principle of feature selection commonly adopted in predictive modeling, which we believe contributes to a more interpretable and potentially more optimal imputation strategy.

Reviewer #3 (Remarks on code availability):

4. The code was made available through a Google drive link. The code itself is not commented and does not come with clear instructions or a README.

Some of the data files (source data and model parameters) as well as function bodies needed to run the code are missing and the results can therefore not be reproduced.

R3-4: Thank you for your feedback. We have substantially updated the code repository by adding more code comments, functions, example data, and a refined README file with clear usage instructions. These materials are included in the updated *SourceCode_KIDS.zip* archive and are also accessible via the following Google Drive link:

<https://drive.google.com/file/d/1MNRjEE1kbOV4S6inqy2BKEn1897cdvX5/view?usp=sharing>.

Zhongshan Ophthalmic Center and The First Affiliated Hospital, Sun Yat-sen University have filed for patent protection for H.L., W.C., Q.W., L.Z., J.L., D.L., and J.W. for work related to non-invasive screening, diagnosis and prognosis prediction of CKD. In line with intellectual property protection policies, detailed model parameters

are not suitable to publicly disclosed at this time. Should the reviewer require access to or have specific inquiries about the model, we kindly encourage direct communication with Prof. Haotian Lin, E-mail: linht5@mail.sysu.edu.cn.

To facilitate practical implementation and independent evaluation, the KIDS system has been deployed on a dedicated cloud platform (<https://zockids.gzzoc.com/modelstore/#!/login>), and an account has been created for reviewer access (Username: USER01; Password: Zoc666). We also prepared a demonstration video to illustrate the system's operational procedures, and example datasets are provided to enable testing and performance assessment of the AI models (included in the updated *SourceCode_KIDS.zip* archive). The online KIDS platform is prepared to be applied to large-scale real-world studies. Continued optimization remains a priority, we are still committed to ensuring methodological transparency and reproducibility through documentation, validation results, and implementation resources-while safeguarding the necessary confidentiality for patent protection.

REVIEWERS' COMMENTS

Reviewer #3 (Remarks to the Author):

I thank the authors for the revised version of the source code and for providing the additional supplementary information on GitHub and as attachments. The added documentation and clarification successfully address the concerns raised in the previous review round.

Minor Comments:

There remains some room for improvement in the codebase. Specifically, I would recommend the following for future refinement:

- Removal of duplicated functions

Response: We have removed the duplicated functions as suggested. We created `ckdfunctions.py` to include the two customized functions and removed the function definition in the other files. The README file has also been updated to reflect this change.

- Reorganization of the file structure for clarity and maintainability

Response: We appreciate your constructive feedback. The file structure has been reorganized to enhance clarity and maintainability.

- Removal of hard-coded parameters and file paths

Response: We have addressed this point by removing hard-coded parameters and file paths to improve code flexibility and reproducibility. The updated code has been submitted as an attachment.

Reviewer #3 (Remarks on code availability):

The code has been made available for review, and documentation was added on the Github repository, in particular through a README.

Missing information from the previous review round was provided, except for the model's parameters which cannot be shared due to intellectual property concerns.